Data Preprocessing & Feature Engineering Report

Adult Income Dataset

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1. Identification of Categorical and Numerical Features

After inspecting the dataset columns and values, features were classified as follows:

Numerical Features

- Age
- Education-num
- Capital-gain
- Capital-loss
- Hours-per-week

These features contain continuous or discrete numeric values suitable for scaling.

Categorical Features

- Workclass
- Education
- Marital-status
- Occupation
- Relationship
- Race
- Sex
- Native-country

The target variable **Income** (<=50K, >50K) is categorical and binary.

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2. Label Encoding for Ordinal Features

Label Encoding was applied to categorical features **where a natural order exists.**

Example

- **Education** (Preschool → Primary → High School → Bachelor → Master → Doctorate)

Reason

- Preserves ordinal relationships.
- Prevents loss of ranking information.
- Suitable for tree-based and linear models.

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3. One-Hot Encoding for Nominal Features

One-Hot Encoding was applied to categorical features **without any inherent order**, such as:

- Workclass
- Occupation
- Marital-status
- Relationship
- Race
- Native-country

Reason

- Prevents misleading ordinal relationships.
- Allows ML models to treat each category independently.
- Improves model interpretability.

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4. Scaling Numerical Features Using StandardScaler

Numerical features were scaled using **StandardScaler**, which transforms values to have:

- Mean = 0
- Standard deviation = 1

Scaled Features

- Age
- Education-num
- Capital-gain
- Capital-loss
- Hours-per-week

Reason

- Ensures all features contribute equally.
- Prevents dominance of large-magnitude values.
- Essential for distance-based and gradient-based algorithms.

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5. Model Readiness: Before vs After Scaling

Aspect	Before Scaling	After Scaling
Feature Range	Highly uneven	Standardized
Model Stability	Lower	Higher
Convergence Speed	Slower	Faster
Algorithm Compatibility	Limited	Broad
ML Readiness	❌ Partial	✅ High

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6. Impact of Scaling on Machine Learning Algorithms

Positive Impact

- Improves performance of:
  - Logistic Regression
  - Support Vector Machines
  - KNN
  - Neural Networks
- Faster convergence during training.
- Better numerical stability.

Minimal Impact

- Tree-based models (Decision Tree, Random Forest) are less affected but still benefit in hybrid pipelines.

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7. Saving the Processed Dataset

After encoding and scaling, the final cleaned dataset was saved as a new file for reuse.

Benefits

- Ensures reproducibility.
- Avoids repeated preprocessing.
- Ready for direct ML model training.

Final Conclusion

The Adult Income dataset initially contained mixed data types and unscaled numerical features, making it unsuitable for direct machine learning. By applying appropriate encoding techniques and feature scaling, the dataset became **fully machine-learning ready**. This preprocessing step significantly improves model performance, fairness, and training efficiency.

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